The Complete Structural Mereology of a Neurosymbolic Search-and-Rescue Manifold: A Unified Axiomatic System Specification

System Architecture Invariant Manifest

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1 Introduction and Architectural Core

We formalize the search-and-rescue plot for a central Fire-Station as an interlocking multi-graph Sequential Dynamic System (SDS) operating across distinct modal boundaries. Following Edward Zalta's Theory of Abstract Objects, the spatiotemporal execution plane maps to ordinary object exemplification (Fx), while the invariant control forests encode properties atemporally (xF). The ultimate objective of the system is the minimization of coordinate uncertainty to realize Mesarovic's Attainable Space (V'), culminating in physical vehicle extraction.

2 The 30-Dimensional Integrated Attribute Matrix

To inject continuous meaning and speech-act context into the message-passing convolutions, the GNN ingests an integrated, multi-scale node attribute matrix $\boldsymbol{X} \in \mathbb{R}^{N \times 30}$:

$$\boldsymbol{x}_i = \left[\boldsymbol{e}_i^{\text{emp (16)}} \mid \boldsymbol{g}_i^{\text{spatial (2)}} \mid \boldsymbol{p}_i^{\text{ppl (3)}} \mid \boldsymbol{r}_i^{\text{rsa (3)}} \mid \boldsymbol{s}_i^{\text{searle (3)}} \mid \boldsymbol{t}_i^{\text{deontic (3)}} \right] \in \mathbb{R}^{30} \tag{1}$$

3 Emergent MARL Utility Sharing and CSMA Protocols

We formalize the coordination of decentralized data center nodes sharing a common electric power grid capacity boundary without centralized top-down scheduling matrices.

3.1 Carrier Sense Multiple Access (CSMA) via Utility Load Sensing

Let $\mathcal{A}_{\text{marl}} = \{a_A, a_B, a_{\text{util}}\}$ represent the decentralized deep reinforcement learning nodes. The aggregate electrical load demand on the substation bus acts as a shared continuous communications carrier channel $L(t)$. Each agent computes its training cycle activation policy $\boldsymbol{\pi}_i$ based on a collective reward framework formulated to minimize spatially correlated active-power disturbances:

$$R_{\text{collective}}(t) = \sum_i \text{Throughload}_i(t) - \gamma \cdot \mathbb{I}\left(\sum_i \text{Compute}_i(t) > L_{\text{capacity}} \right) \tag{2}$$

where $\mathbb{I}$ represents the indicator function tracking synchronized load overlaps, and $\gamma = 5000$ serves as a non-linear penalty multiplier. If an agent senses a busy carrier state ($L(t) \geq 0.60$), the local

policy triggers an automatic exponential backoff routine:

$$\boldsymbol{\pi}_i(t) \implies \text{Backoff}(t + \Delta t_{\text{random}}) \tag{3}$$

releasing channel capacity natively without explicit inter-company planning loops.

3.2 Topological Load Perspective Tracing under Synchronous Attack

When an external adversary or malicious perimeter source executes a synchronous load intrusion ($\Delta L_{\text{attack}} = +0.85$ GW), the MARL network initiates a distributed geometric trace query. The load anomaly is verified by evaluating localized structural load perspectives $\mathcal{P}_k$ across the multi-graph substation topology:

$$\nabla_{\boldsymbol{x}} V_k(\mathcal{P}_k) \neq 0 \implies \text{Isolate}(\text{Branch}_{\text{unmapped}}) \tag{4}$$

If the gradient reveals that the load advection path originates from an unmapped physical branch rather than a registered consumer node, the attention guardrail issues a hard `activate_mission_abort_safety_hal` token, freezing the transmission bus to protect structural grid connectivity.